COMSATS University Islamabad, Lahore campus

Assignment #1

Software Design And Architecture

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Section: A

SOFTWARE DESIGN AND ARCHITECTURE ASSIGNMENT

System Selected: Uber Eats

1. Introduction

Modern online food delivery platforms let users order food through mobile or web applications and have it delivered to their location. One of the most popular platforms for this is **Uber Eats.**

Uber Eats connects three main groups:

• Customers

• Restaurants

• Delivery partners

The system handles restaurant listings, order placement, payment processing, and delivery tracking.

Because of the complexity of this system, software architecture is crucial for organizing it into manageable parts. This assignment looks at Uber Eats using the software architectural styles and patterns covered in the course.

2. Architectural Style Used

The best architectural styles for Uber Eats are:

1. Layered Architecture

The system is divided into layers, and each layer provides services to the layer above it.

Typical layers include:

- Presentation Layer
- Business Logic Layer
- Data Layer

This separation improves **maintainability, scalability, and testability**.

2. Client–Server Architecture

Uber Eats operates on a **client-server model** where:

- Clients → Mobile apps / web apps used by customers, restaurants, and drivers
- Server → Backend services that process requests and manage data

The client sends requests, like placing an order, and the server processes these requests and returns results.

3. System Modules of Uber Eats

There are many parts to the Uber Eats system, each of which does a different job.

1. Module for managing users

Handles signing up, logging in, and verifying users.

What it does:

- Make accounts for users
- Verifying login
- Managing profiles

2. Module for Managing Restaurants

Lets restaurants control how they show up on the site.

What it does:

- Managing the menu
- Lists of restaurants
- Accepting orders

3. Module for Managing Orders

Takes care of every step of a food order.

What it does:

- Placing an order

- Confirm your order
- Following your order
- Finish the order

4. Module for Processing Payments

In charge of processing payments.

What it does:

- Paying online
- Checking the validity of transactions
- Confirmation of payment

5. Module for Managing Delivery

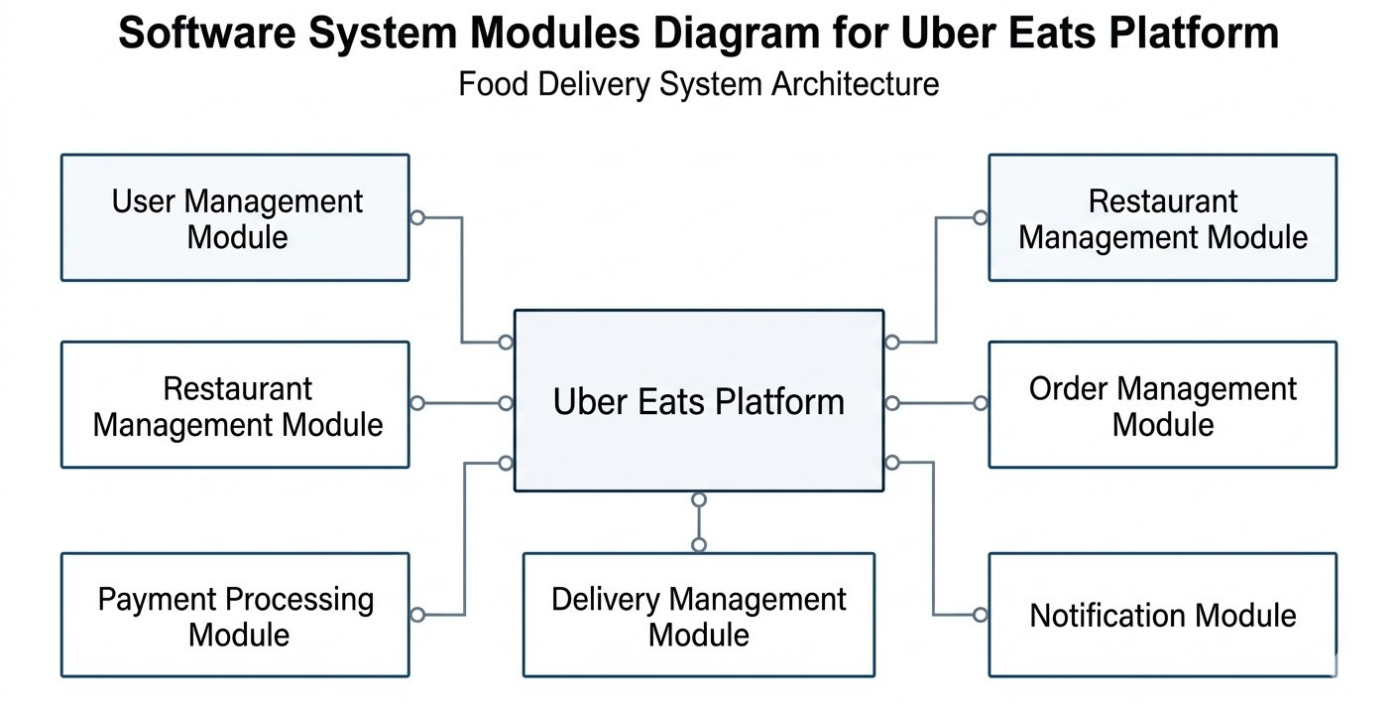
Manages drivers and keeps track of their routes.

What it does:

- Give drivers their jobs
- Keep an eye on where the package is being delivered
- Change the status of the delivery

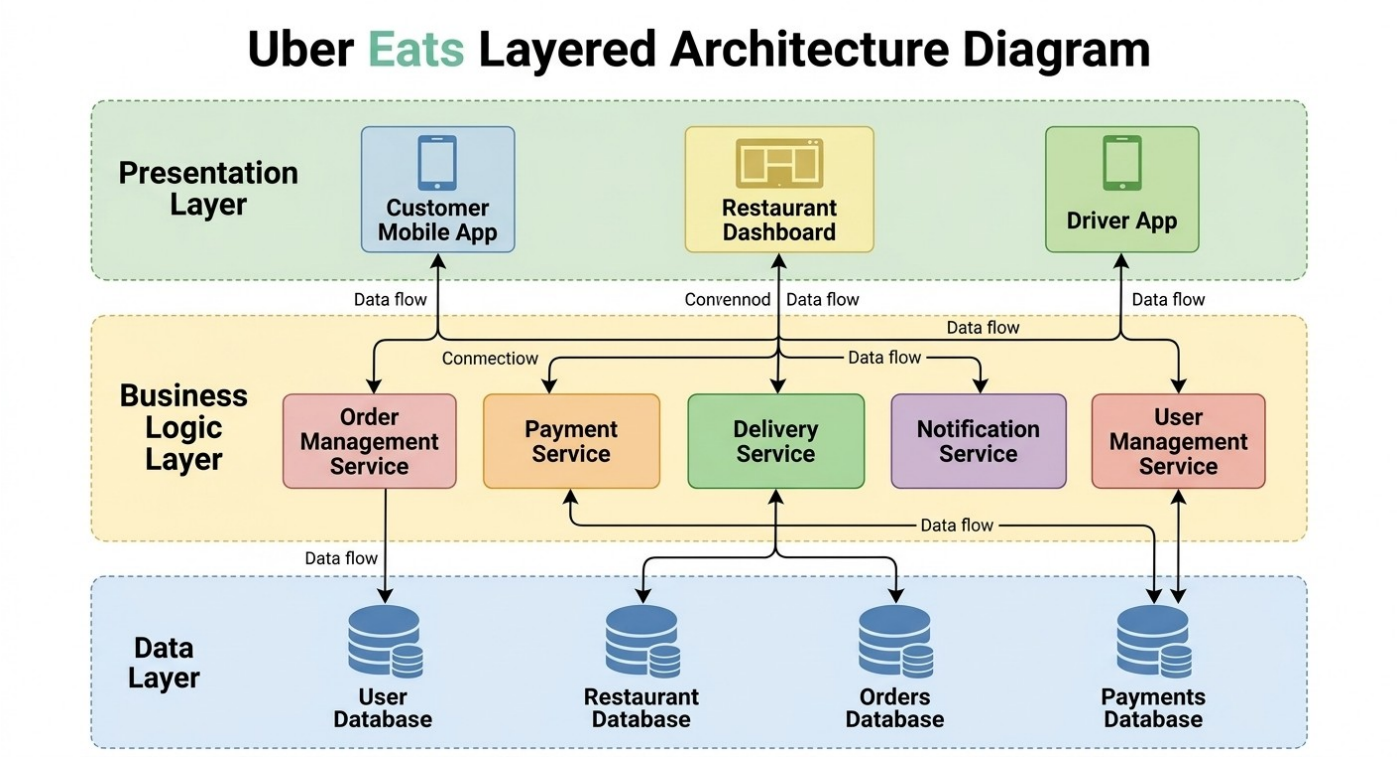
6. Module for notifications

Sends alerts in real time.
For example:
- Order confirmed
- Driver given
- Food will be here soon



4. Uber Eats System Architecture

Uber Eats follows a **Layered Architecture consisting of three main layers**.



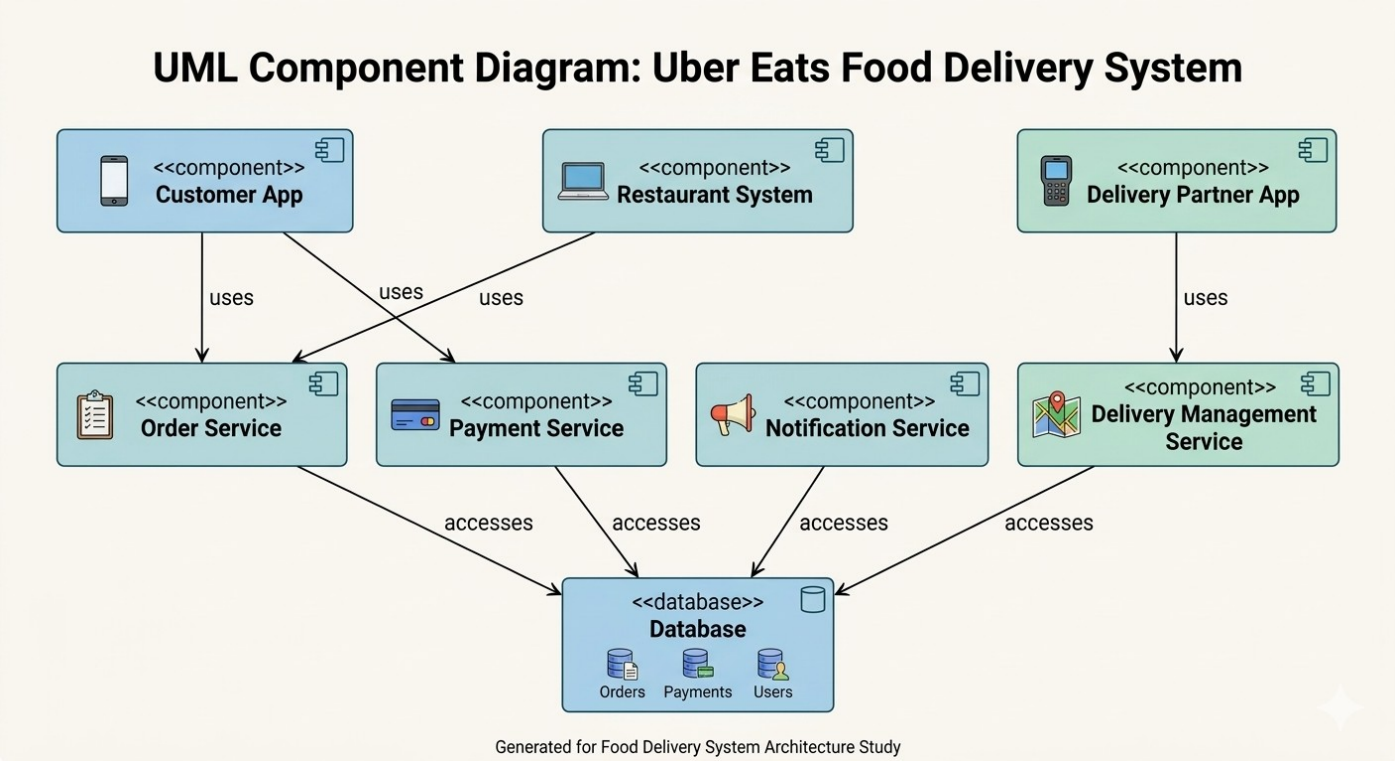
5. Components of the System

Components are the major building blocks of the system.

Main components include:

Component	Description
Customer App	Interface used by the customer to place orders for food
Restaurant System	Allows the restaurant to manage orders and menus
Delivery Partner App	Used by the driver to manage orders
Order Service	Handles the creation of orders and processing of orders
Payment Service	Handles transaction management

Notification Service	Handles sending of notifications and updates
Database	Handles storing of the system data



6. Connectors

The connectors describe how the components in the system communicate with each other.

The connectors of the Uber Eats ordering system include:

1. API Calls

The mobile application communicates with the backend using REST API calls.

Example:
Customer App → Order Service

2. Network Communication

The clients communicate with the servers using the internet using HTTP/HTTPS.

3. Database Queries

The business logic layer uses SQL queries to communicate with the database.

Example:
Order Service → Orders Database

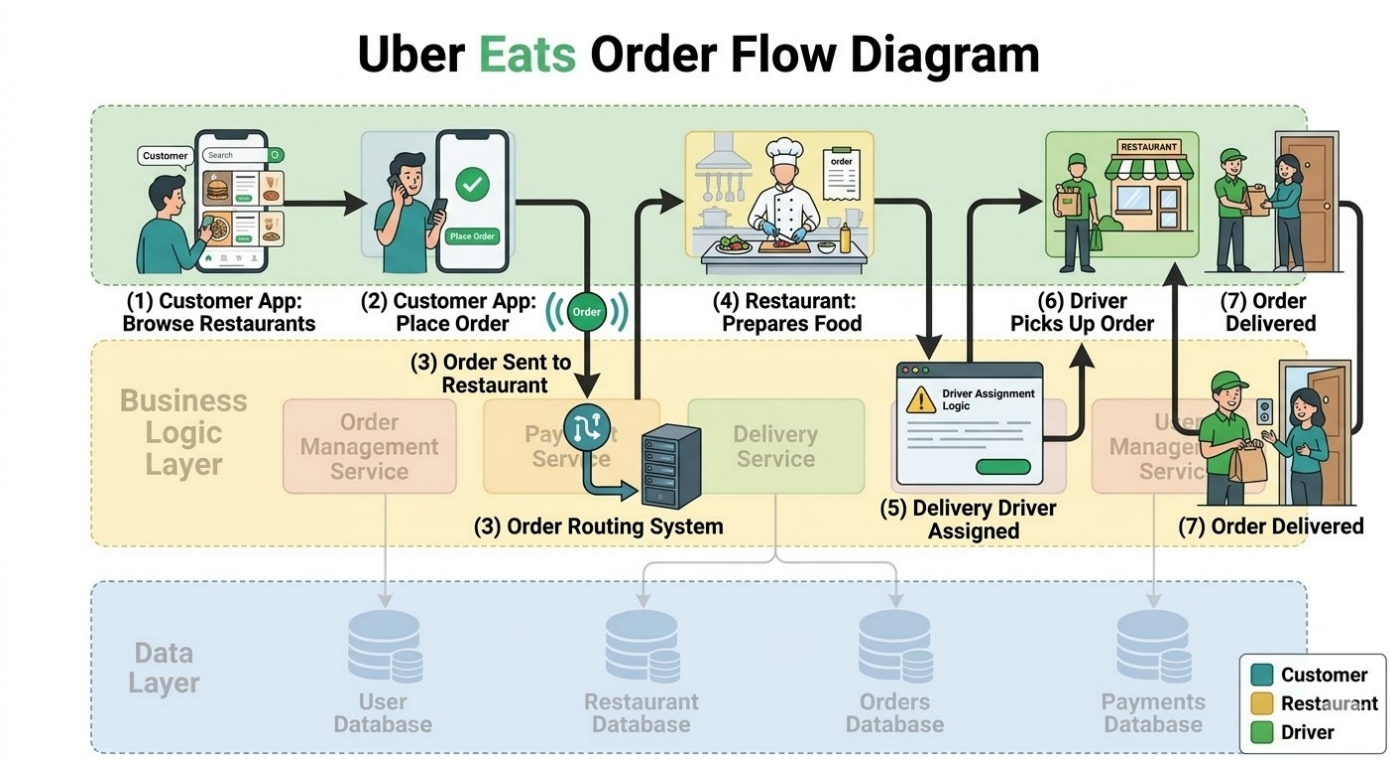
7. Data Elements

The data elements describe the information that is moving around in the system.

The most significant data elements include:

Data Elements Table

User Data	It has details about customers like their name, phone number, email address, and delivery address.
Restaurant Data	Includes information about the restaurant, such as its name, location, menu items, and prices.
Order Data	Keeps track of information about orders, like the order ID, the food items chosen, the number of items, and the status of the order.
Payment Data	Transaction information like Transaction ID, payment method (card, wallet, etc.), and payment status are all included.
Delivery Data	Includes information about the delivery, such as the driver's name, location, and status.



8. Topology

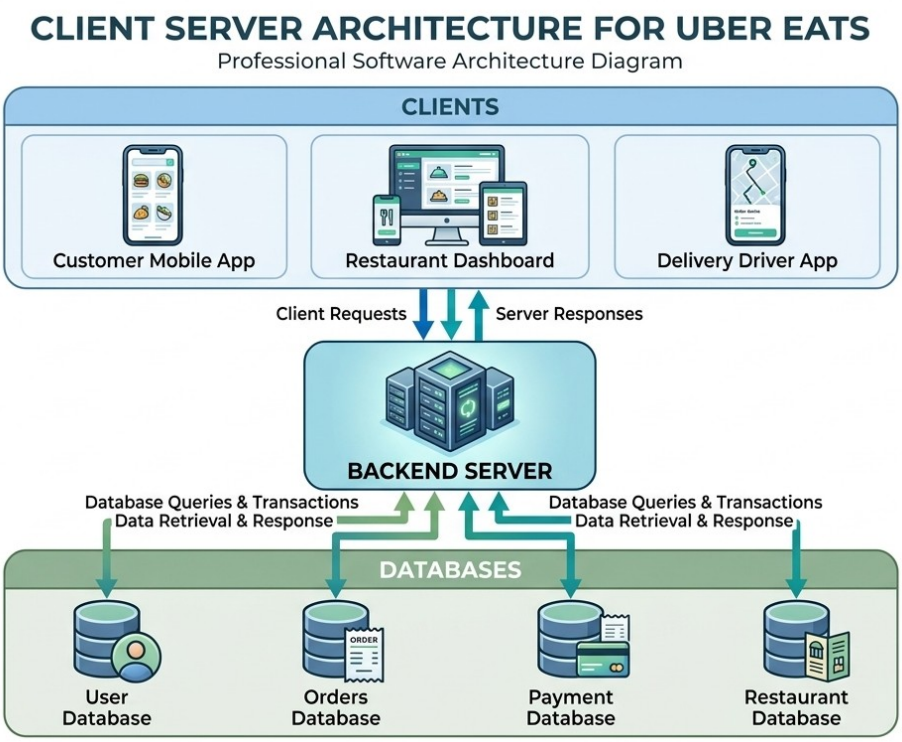
Topology studies how different parts are connected.

The Client–Server Topology is what Uber Eats uses.

Here's how it all fits together:

1. Customers use an app.
2. Restaurants have their own system.
3. Drivers also have an app. All of these clients communicate with a central backend server.
4. This server manages incoming requests and pulls data from the databases.

All clients talk to the central backend server, which handles requests and gets data from databases.



9. Constraints Imposed by the Architecture

Certain rules must be followed in the layered architecture.

Examples:

1. Presentation layer cannot directly access the database.
2. All requests must pass through the business logic layer.
3. Each module must communicate using defined APIs.

These constraints maintain **system organization and modularity**.

10. Qualities Yielded by the Architecture

The architecture provides several software quality attributes.

1. Scalability

The system can handle many users simultaneously.

2. Maintainability

Each module can be modified independently.

3. Reusability

Services like payment and notifications can be reused.

4. Security

Sensitive operations like payments are handled in secure layers.

5. Testability

Each layer can be tested independently.

11. Typical Uses

You can often find layered and client-server architectures in:

- E-commerce systems
- Banking applications
- Online reservation systems
- Food distribution platforms

  These designs are effective for systems with a lot of customers using the same core services.

Although the architecture has many benefits, it also has some disadvantages.

1. Performance Overhead

Requests have to go through a lot of layers, which could make the response time longer.

2. More Complicated

A lot of modules and services may be needed for big systems.

3. Difficulty of Deployment

When business logic changes, it might be necessary to make changes in more than one layer.

13. Conclusion

Uber Eats connects consumers, restaurants, and delivery partners in a sophisticated network. Because Layered Architecture and Client-Server Architecture are used, the system is flexible, modular, and simple to maintain.

The three-layer architecture helps developers manage complexity and create big, dependable systems more easily.